



Comsats University Islamabad, Lahore Campus

Course Title:	DevOps for Cloud Computing				Course Code:	CSC331	Credit Hours:	(1)
Course Instructor:	M Khizar Hayat				Program Name:	BSCS		
Semester:	6 th	Batch:	FALL 2023	Section:	C	Date:	29-04-2026	
Due Date:	30-04-2026,				Maximum Marks:		20	
Student's Name:					Reg. No.			

OBJECTIVES

Students will design and implement an industry-aligned for a static website that enforces code quality, automated builds, and containerization using Docker.

By the end of this assignment, students will be able to:

- Design YAML-based GitHub Actions workflows.
- Enforce HTML and CSS quality using HTMLHint and Stylelint.
- Build static websites using Parcel inside CI.
- Create Dockerfiles and build Docker images.
- Collaborate using feature branches, PRs, and branch protection rules.

IMPORTANT INSTRUCTIONS

1. Group Size: Five (5) students per group.
2. Team Lead (TL): One student will act as the Team Lead, the other four as Team Members.
3. Repository Setup: TL must create a NEW GitHub repository
4. Manual Docker Commands: Manual docker. Docker images must be built and pushed only through GitHub.
5. Submission: Only the Team Lead will submit this document.
6. Plagiarism: Identical repositories, workflows, commits, or Dockerfiles will result in ZERO marks.

Table 1: Group Details

Sr #	Student	GitHub	Role	Branch	Verification
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	Name	Username	(Lead/Member)	Worked On	Screenshots Added (Yes/No)
1					
2					
3					
4					
5					

Group Repository URL: [Paste Your GitHub Repository URL Here]

SCENARIO

University Digital Notice Board

Your team will collaboratively develop a University Digital Notice Board static website that displays academic announcements, examination schedules, and admission notices.

The website MUST pass HTMLHint, Stylelint, and Parcel build stages in GitHub Actions BEFORE a Docker image is allowed to be built and pushed.

REQUIRED FOLDER STRUCTURE

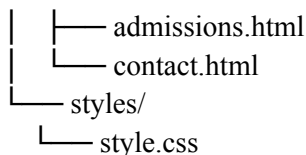
uni-notice-board/

```
|— .github/
|   |— workflows/
|   |— ci.yml
```

```
|— .gitignore
|— .dockerignore
|— Dockerfile
|— package.json
|— README.md
|— src/
|   |— index.html
|   |— notices.html
|   |— exams.html
```



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PAGE ASSIGNMENTS

Page	Description	Responsibility
index.html	Homepage & navigation	Team Lead
notices.html	Academic notices	Member 1
exams.html	Exam schedules	Member 2
admissions.html	Admission updates	Member 3
contact.html	Contact details	Member 4

CI PIPELINE (MANDATORY SEQUENCE)

- 1: Checkout Code
- 2: HTMLHint ()
- 3: Stylelint ()
- 4: Build Website using Parcel
- 5: Build Docker Image using Dockerfile
- 6: Push Docker Image to Docker Hub

NOTE: Create three separate jobs in workflow.

CRITICAL WORKFLOW RULES (MANDATORY)

- EACH team member MUST locally run:
 - HTMLHint
 - Stylelint
 - Parcel build
 - Docker image build
 - Docker container run
- Screenshots of ALL above steps MUST be pasted at the END of this document by EACH member.
- NO commits are allowed before successful local verification.
- Each member must push a feature branch and create a Pull Request (PR).



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- Team Lead MUST create a Branch Protection Rule on the 'develop' branch.
- PRs MUST NOT be merged by team members. ONLY Team Lead is allowed to merge PRs.

TASKS & MARKING

1. Repository initialization & collaborators
2. Folder structure & Dockerfile creation
3. Branching strategy & protection rules
4. HTMLHint integration
5. Stylelint integration
6. Docker image build
7. Docker Hub push
8. Local verification screenshots
9. Successful run

SUBMISSION REQUIREMENTS

The Team Lead must submit:

- This completed document
- GitHub repository URL
- Docker Hub repository link
- Screenshots section completed by ALL members
- Short reflection (3–4 lines)